

UNICON: THE ANONYMOUS UNIVERSITY COMMUNICATION PLATFORM

A PROJECT REPORT
BY

TEAM NO. 5

SUMANT DESWAL (S24CSEU2358)
VAIBHAV YADAV (S24CSEU2340)
AADARSH DANGI (S24CSEU2332)
PARTH SHARMA (S24CSEU2370)



SUBMITTED TO

SCHOOL OF COMPUTER SCIENCE ENGINEERING AND
TECHNOLOGY, BENNETT UNIVERSITY

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DECLARATION

I/We hereby declare that the work which is being presented in the report entitled “UniCon: The Anonymous University Connection Platform”, is an authentic record of my/our own work carried out during the period from JAN, 2026 to April, 2026 at School of Computer Science and Engineering and Technology, Bennett University Greater Noida.

The matters and the results presented in this report has not been submitted by me/us for the award of any other degree elsewhere.

Signature of Candidate

Sumant Deswal
(Enroll. No. S24CSEU2358)

Vaibhav Yadav
(Enroll. No. S24CSEU2340)

Aadarsh Dangi
(Enroll. No. S24CSEU2332)

Parth Sharma
(Enroll. No. S24CSEU2370)

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Signature of Candidate

Sumant Deswal

(Enroll. No. S24CSEU2358)

Vaibhav Yadav

(Enroll. No. S24CSEU2340)

Aadarsh Dangi

(Enroll. No. S24CSEU2332)

Parth Sharma

(Enroll. No. S24CSEU2370)

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ABSTRACT

Unicon is a social media platform designed by keeping in mind the needs and requirements of building a safe community where students can interact with each other and find people to connect with. It maintains a layer of anonymity by allowing students to post about their interests anonymously. Posts can be filtered using tags to maintain relevancy and allowing students to choose what they're interested in. Accountability is important when it comes to anonymity so we will have moderators on the site with access to user's personal information such as college email to take appropriate actions upon misuse of the platform. Students will be able to message each other and make friends through the platform, making the university experience more comfortable and friendly.

1. INTRODUCTION

In recent years, digital communication platforms have become an essential part of student life, enabling collaboration, discussion, and knowledge sharing. However, most existing platforms lack anonymity, which often discourages students from expressing their genuine thoughts, asking questions, or discussing sensitive topics. Fear of judgment, social pressure, and identity exposure limit open communication within university communities.

UniCon is designed to address this gap by providing a secure and anonymous platform exclusively for university students. The platform enables users to share ideas, participate in discussions, and connect with peers without revealing their identities. By combining anonymity with structured interaction through tags, profiles, and moderation, the system promotes a balanced environment of freedom and responsibility.

The platform integrates multiple features such as anonymous posting, categorized discussions, private messaging, and a friend system. These features aim to enhance engagement while maintaining user safety. UniCon ultimately fosters a collaborative and inclusive digital ecosystem within academic institutions.

1.1. Problem Statement

Currently, students lack a dedicated platform where they can freely express opinions, seek help, and engage in discussions without fear of judgment or identity exposure. Existing social media platforms are either not anonymous or not tailored specifically for university communities, resulting in limited meaningful interaction and reduced participation in open discussions.

2. BACKGROUND RESEARCH

2.1. Proposed System

UniCon is a web-based anonymous communication platform designed specifically for university students. The system allows users to create anonymous profiles and participate in discussions across various topics such as academics, sports, technology, and entertainment.

The platform provides a structured environment where users can post content anonymously, interact through comments, and connect privately via direct messaging. A tagging system ensures organized content discovery, while a friend system enables users to build connections beyond public discussions.

To maintain a safe and respectful environment, moderation controls are implemented. Administrators can access user information only in cases of misuse or violations, ensuring accountability without compromising anonymity.

The vision of the project is to create a secure, inclusive, and engaging digital space where students can communicate freely and build meaningful connections.

2.2. Goals and Objectives

Table 1: Goal and Objectives

#	Goal or Objective
1	Enable anonymous communication among students
2	Provide a secure moderated platform by maintaining balance between anonymity and accountability
3	Ensure easy to use intuitive UI
4	Support categorized discussions using tags and implement private messaging
5	Have fun working on the project

3. PROJECT PLANNING

3.1. Project Lifecycle

The project follows an **Agile development methodology**. The development process was divided into multiple iterations, where features such as authentication, posting system, messaging,

and moderation were implemented incrementally. Regular testing and feedback were incorporated during each phase to improve system performance and usability.

3.2. Project Setup

The UniCon project involved several important technical and development-related decisions to ensure scalability, maintainability, and efficient collaboration among team members.

Firstly, the development environment and technology stack were carefully selected. The project was implemented as a web-based application using modern web technologies. The frontend was developed using responsive design principles to ensure accessibility across devices, while the backend was designed to handle user authentication, data processing, and communication efficiently. A database system was chosen to securely store user data, posts, messages, and interactions.

For version control and collaboration, Git was used along with GitHub as the central repository. This allowed team members to work simultaneously on different features, manage code versions, and track changes effectively. Proper branching strategies were followed to separate development and stable code.

Coding standards and best practices were maintained throughout the project to ensure readability and maintainability of the codebase. Modular design principles were followed so that different components such as authentication, posting system, messaging, and user management could be developed and tested independently.

Security was a key consideration in UniCon. Since the platform is based on anonymity, decisions were made to restrict public exposure of personal data while still allowing moderators controlled access in case of misuse. Input validation and basic security measures were implemented to prevent malicious activities.

Additionally, the project was designed to be scalable and extensible. Future features such as mobile support, advanced moderation, and AI-based filtering can be integrated without major restructuring of the system.

Overall, these decisions ensured that UniCon is robust, user-friendly, and adaptable for future enhancements.

3.3. Stakeholders

Table 2:

Stakeholder	Role
Sumant Deswal	Team Leader
Dr. Chandra Bhushan	Mentor
Vaibhav Yadav	Team member
Aadarsh Dangi	Team member
Parth Sharma	Team member

3.4. Project Resources

<< Identify the anticipated resources required for this project. This can include staff members who will work on the project, equipment needed for the project, special software that will need to be acquired, or any other resource necessary for the project. >>

Example:

Table 3: Sample 4

Resource	Resource Description	Quantity
Database Server	MongoDB was used	1
The Team	Our team of students who will be the primary developers of the project.	4
Mentor	The mentor who will be able to provide us with technical assistance.	1
Windows PC	A windows PC with a browser to be used for testing the app and the entire development	2

3.5. Assumptions

Table 4:

#	Assumption
A1	The capstone team and mentors will be able to meet face to face once a week.
A2	All team members have access to the required development tools and a stable internet connection
A3	The GitHub repository will remain accessible for version control and collaboration throughout the project
A4	Team will have sufficient time to complete a working model to present by mid-semester
A5	The database and server infrastructure will be available and function reliably during development and testing
A6	Basic security measures implemented will be sufficient for initial deployment
A7	The system will primarily be accessed through web browsers on laptops or mobile devices

4. PROJECT TRACKING

4.1. Tracking

The progress of the UniCon project was continuously monitored using version control systems and collaborative tools. Git was used as the primary version control system, with GitHub serving as the central repository for storing and managing the project code. This enabled efficient collaboration among team members, allowing them to work on different modules simultaneously while maintaining code integrity.

Each feature of the system, such as authentication, posting, messaging, and user management, was developed in separate branches and later merged into the main branch after testing. Regular commits were made to track progress and ensure that changes were documented properly.

Bug tracking and issue management were handled using GitHub Issues, where team members reported bugs, suggested improvements, and tracked pending tasks. This helped in maintaining transparency and accountability within the team.

Project documentation, including reports and presentation materials, was also maintained alongside the code repository for easy access and version management. Regular testing was conducted to ensure that newly added features did not affect existing functionality.

Example:

Table 5: Sample 6

Information	Description	Link
Code Storage	Project code will be stored in GitHub repository.	Link
Project Documents and Assignments	The documentation is stored in the GitHub repository	Link

4.2. Communication Plan

Effective communication was essential for the successful development of the UniCon platform. The team established a structured communication plan to ensure coordination among team members, mentor, and other stakeholders. Regular meetings and updates were conducted to track progress, resolve issues, and align development goals.

Table 6: Regularly Scheduled Meetings

Meeting Type	Frequency/Schedule	Who Attends
Team Meeting	Weekly	Project team
Mentor Meeting	Weekly in class	Project team, Mentor
Progress Meeting	Every 3 days	Some group members

Table 7: Information To Be Shared Within Our Group

Who?	What Information?	When?	How?
Project team	Task assignments & General scrum information	Weekly	Team meetings, listing in Project Specification.

Table 8: Information To Be Provided To Other Groups

Who?	What Information?	When?	How?
Mentor	Final deliverables	At completion of project	Project specification doc., code, Power Point presentation
Mentor	Weekly progress report	Weekly	In Class

Table 9: Information Needed From Other Groups

Who?	What Information?	When?	How?
Mentor	Requirement changes	Every week	In Class

4.3. Deliverables

The UniCon project is expected to produce several key deliverables that demonstrate the design, implementation, and functionality of the system. These deliverables include both technical outputs and documentation that reflect the development process and final outcome of the project

Table 10: Deliverables

#	Deliverable
1	Functional Platform
2	Source code with documentation
3	User Manual

4	PPT explaining the project
5	Future scope and enhancement plan
6	Final report (final PowerPoint presentation, 5 minute video, and final sprint)

5. SYSTEM ANALYSIS AND DESIGN

This section describes in detail about the design part of the system.

5.1. Overall Description

UniCon is a web-based anonymous communication platform developed specifically for university students to facilitate open discussions, peer interaction, and community engagement within a secure environment. The primary objective of the system is to eliminate the fear of judgment and identity exposure that often restricts students from expressing their thoughts freely on traditional social media platforms.

The platform enables users to create anonymous profiles with limited contextual identifiers such as course, academic year, and residential status. This approach ensures that discussions remain relevant while preserving user privacy. Students can create posts, participate in discussions through comments, and explore content categorized using tags such as academics, sports, technology, and entertainment. The tagging system enhances content organization and allows users to easily discover topics of interest.

UniCon also includes a private messaging feature that allows users to communicate one-on-one. While interactions begin anonymously, users have the flexibility to share their identity voluntarily, enabling a transition from anonymous to personal communication when desired. Additionally, the platform incorporates a friend system that allows users to connect and maintain ongoing interactions beyond public discussions.

From a technical perspective, the system follows a modular architecture where different components such as authentication, posting, messaging, and user management operate independently but are integrated to provide a seamless user experience. The backend handles data processing, storage, and communication, while the frontend ensures an intuitive and responsive interface.

To maintain a safe and respectful environment, UniCon implements moderation controls. Administrators have restricted access to user data and can intervene only in cases of misuse or policy violations. This ensures a balance between anonymity and accountability.

Overall, UniCon provides a scalable, user-friendly, and secure platform that encourages honest expression and meaningful interaction among students.

5.2. Users and Roles

The UniCon platform consists of different types of users, each with specific roles and responsibilities. These roles ensure proper functioning of the system while maintaining a balance between anonymity, usability, and security.

Table 11:

User	Description
Developer	A team member who is going to be working on the frontend and backend code
Bug Tester	A team member who actively tests the platform for any bugs and reports back.
Moderator	A person who will ensure the moderation of the platform and prevent misuse

5.3. Design diagrams/Architecture/ UML diagrams/ Flow Charts/ E-R diagrams

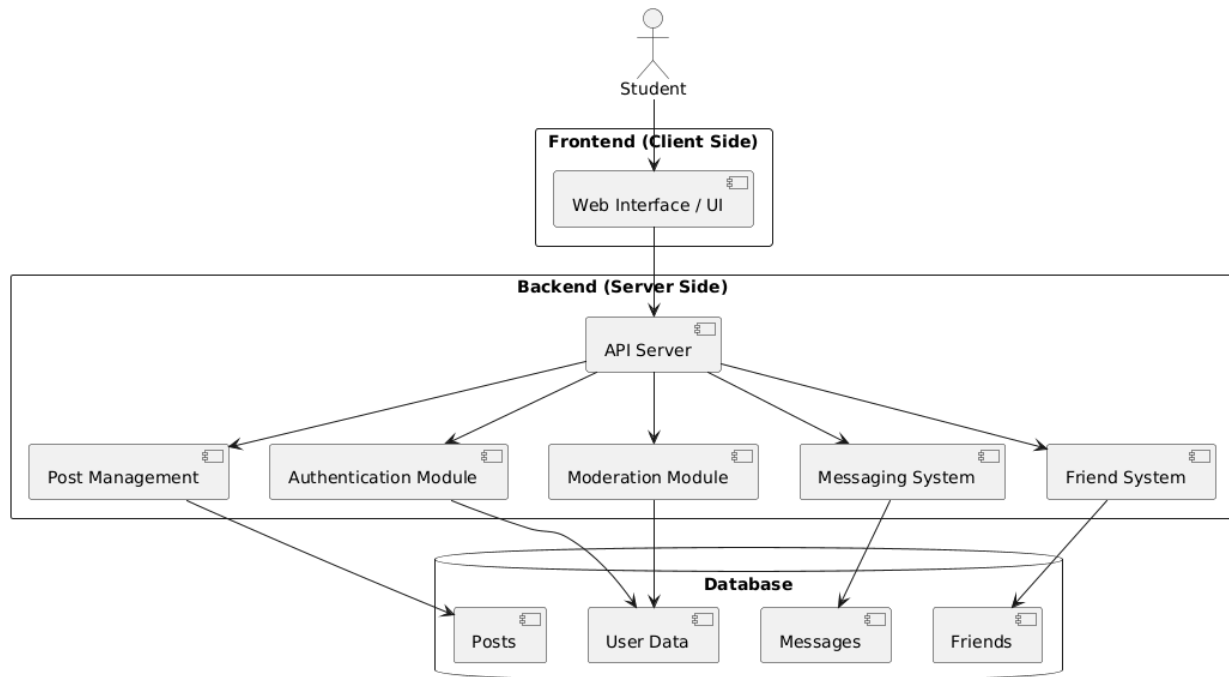
5.3.1. Product Backlog Items

- As a **student**, I want to register and log in so that I can access the platform securely.
- As a **student**, I want to create an anonymous profile so that my identity remains hidden while interacting.
- As a **student**, I want to create posts so that I can share my thoughts and ideas.
- As a **student**, I want to view posts from other users so that I can stay updated with discussions.
- As a **student**, I want to comment on posts so that I can participate in discussions.
- As a **student**, I want to like or interact with posts so that I can express my opinion.
- As a **student**, I want to filter posts using tags so that I can find relevant content easily.
- As a **student**, I want to send and receive private messages so that I can communicate personally with others.
- As a **student**, I want to add or remove friends so that I can maintain connections.
- As a **student**, I want to view my profile so that I can track my activity.
- As a **moderator**, I want to monitor posts so that inappropriate content can be identified.
- As a **moderator**, I want to review reported content so that necessary action can be taken.

- As a **moderator**, I want to remove inappropriate posts so that the platform remains safe and respectful.

5.3.2. Architecture Diagram

The UniCon system follows a client-server architecture where the frontend interacts with backend services through APIs. The backend processes user requests, handles authentication, and communicates with the database for storing and retrieving data. The system is designed in a modular way to ensure scalability and maintainability.



5.3.3. Use Case Diagram

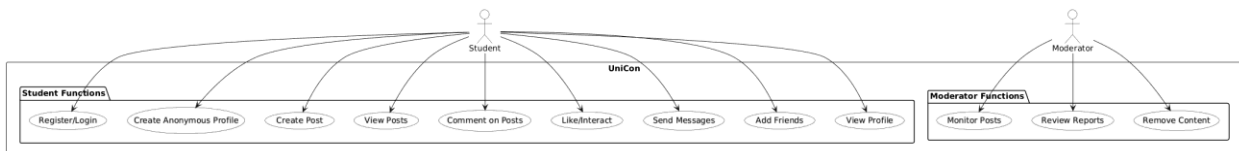


Figure 1: Use Case Diagram

5.3.4. Class Diagram

The class diagram of UniCon represents the structure of the system by illustrating the main classes, their attributes, methods, and relationships. It includes core entities such as User, Post, Comment, Message, and Friend, along with their interactions. The diagram reflects how different components of the platform are interconnected to support functionalities like posting, messaging, and moderation.

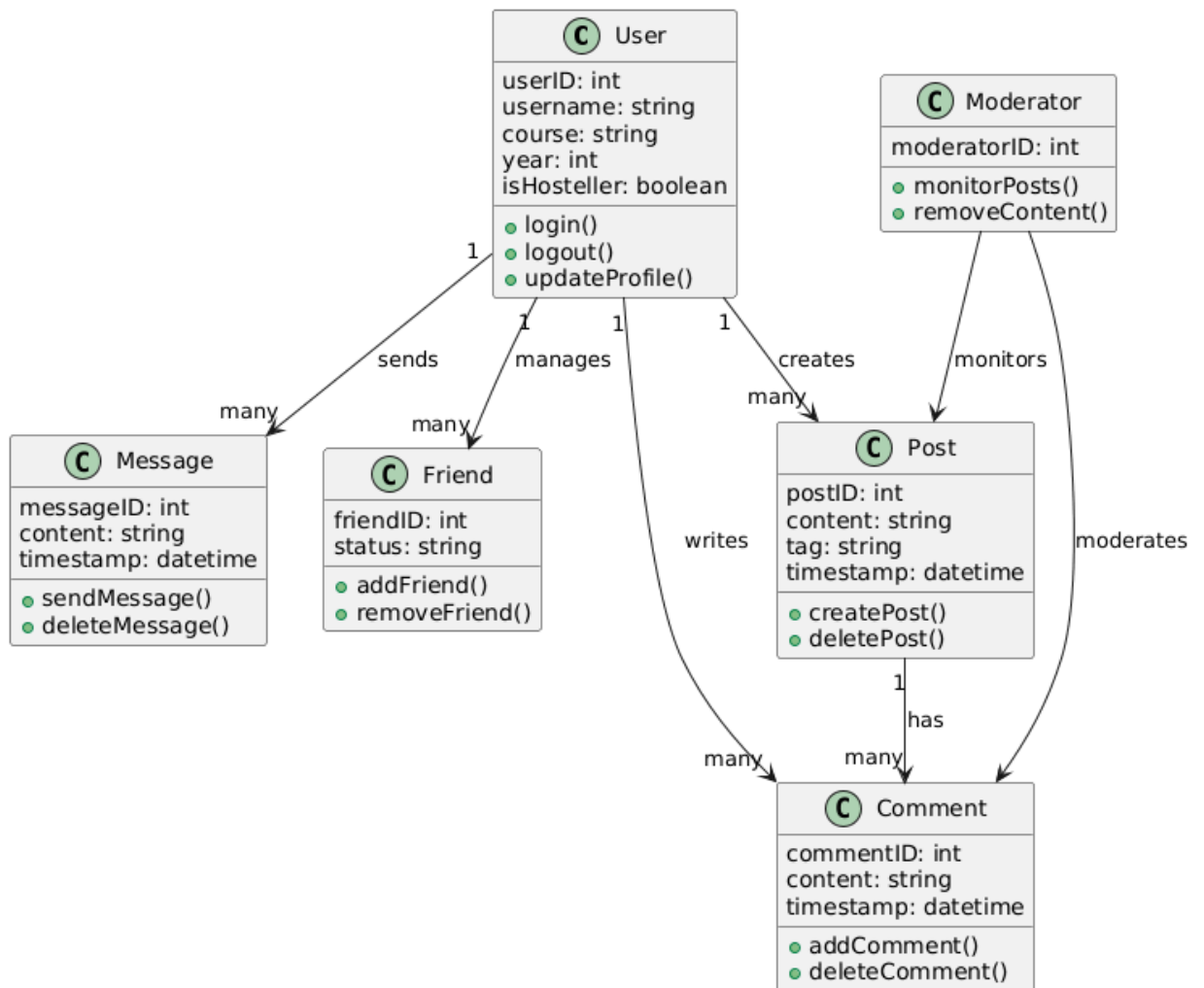


Figure 2

5.3.5. Activity Diagrams

The activity diagram represents the workflow of a user interacting with the UniCon platform. It illustrates the sequence of actions from logging into the system to performing activities such as creating posts, interacting with content, sending messages, and logging out. The diagram helps visualize the overall user journey and system behavior.

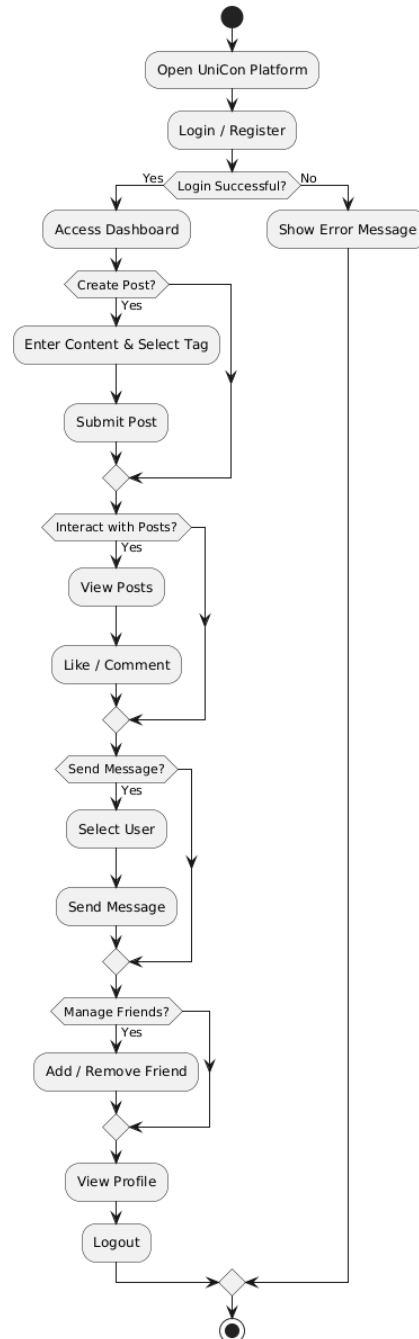


Figure 3

5.3.6. Sequence Diagram

The sequence diagram illustrates the interaction between different components of the UniCon system during the process of creating a post. It shows how the user communicates with the frontend, which then interacts with the backend server and database to store and display the post. The diagram highlights the flow of data and control between system components.

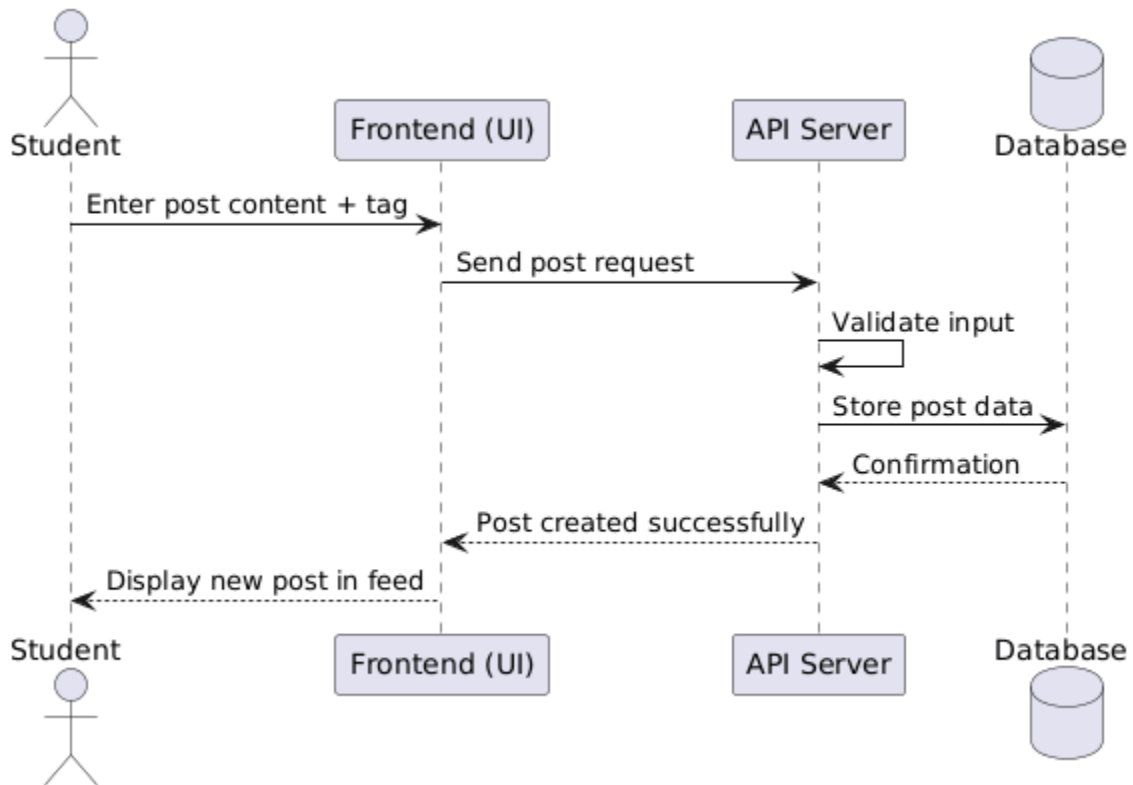


Figure 4

5.3.7. Data Architecture

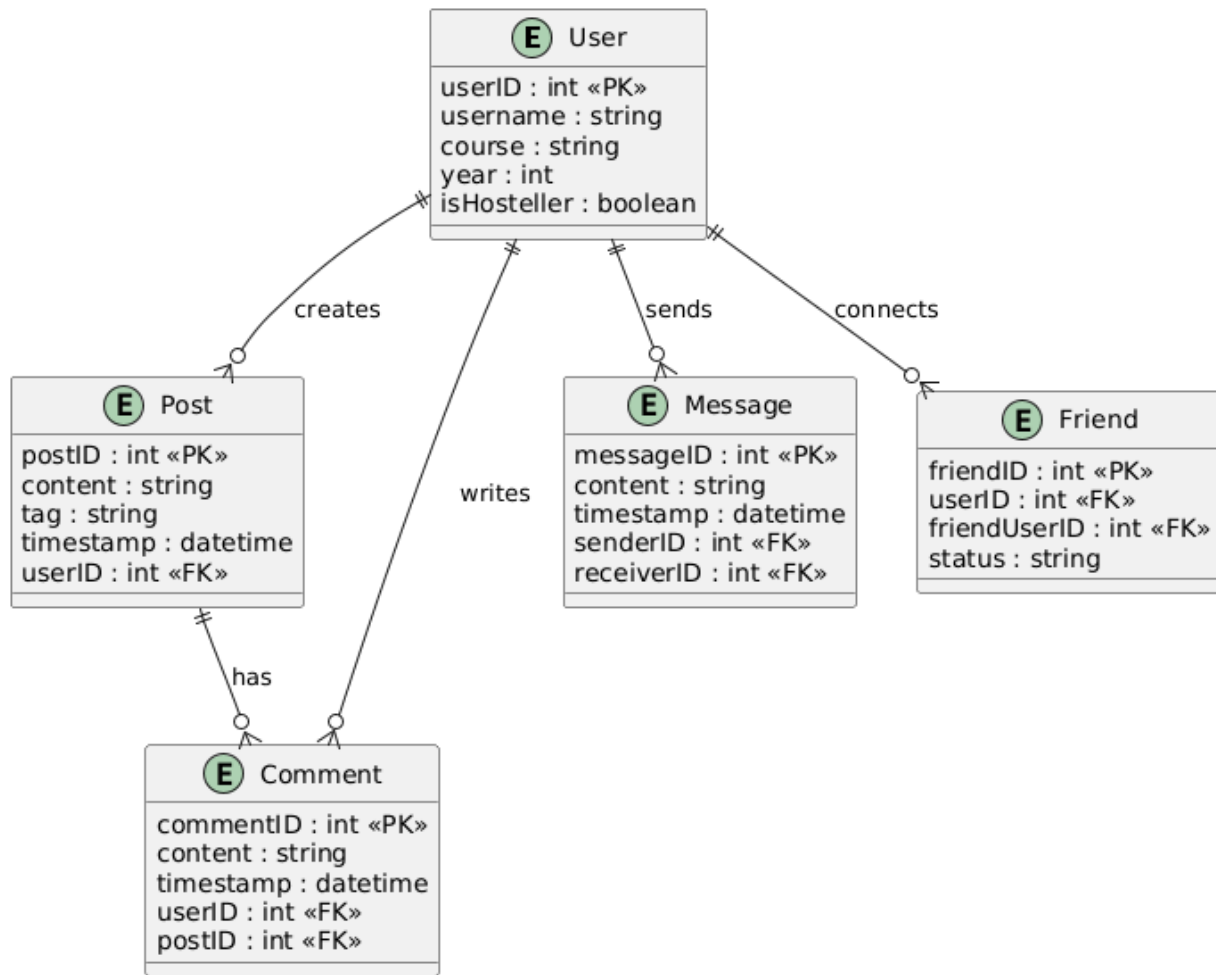


Figure 5

6. USER INTERFACE

6.1. UI Description

The UniCon platform features a clean, modern, and user-friendly interface designed to provide a seamless experience for university students. The interface follows a minimalistic design approach with a consistent color scheme and intuitive navigation, ensuring ease of use even for first-time users.

The **landing page** presents the core idea of the platform with a clear tagline and options for login and registration. It provides a simple entry point for users to access the system without unnecessary complexity.

Once logged in, users are directed to the **dashboard**, which serves as the central hub of the platform. The dashboard displays a feed of posts created by other users. Each post includes content, tags, and interaction options such as commenting and engaging with posts. A post creation section is prominently placed, allowing users to quickly share their thoughts by entering content and selecting relevant tags.

The interface includes a **tagging system**, where posts are categorized into topics such as academics, technology, social discussions, and entertainment. This helps users easily navigate and find relevant content.

The **friends page** allows users to search for other users, send friend requests, and manage their connections. The design is simple and organized, making it easy to track friend relationships.

The **messaging interface** provides a clean chat layout where users can send and receive private messages. Conversations are displayed in a structured format, ensuring smooth communication between users.

The **profile page** displays user-related information such as activity, posts, and interactions while maintaining anonymity. It provides users with an overview of their engagement on the platform.

Overall, the UniCon UI is designed to be responsive, accessible, and intuitive, enabling users to interact efficiently while maintaining anonymity and focus on content.

6.2. UI Mockup

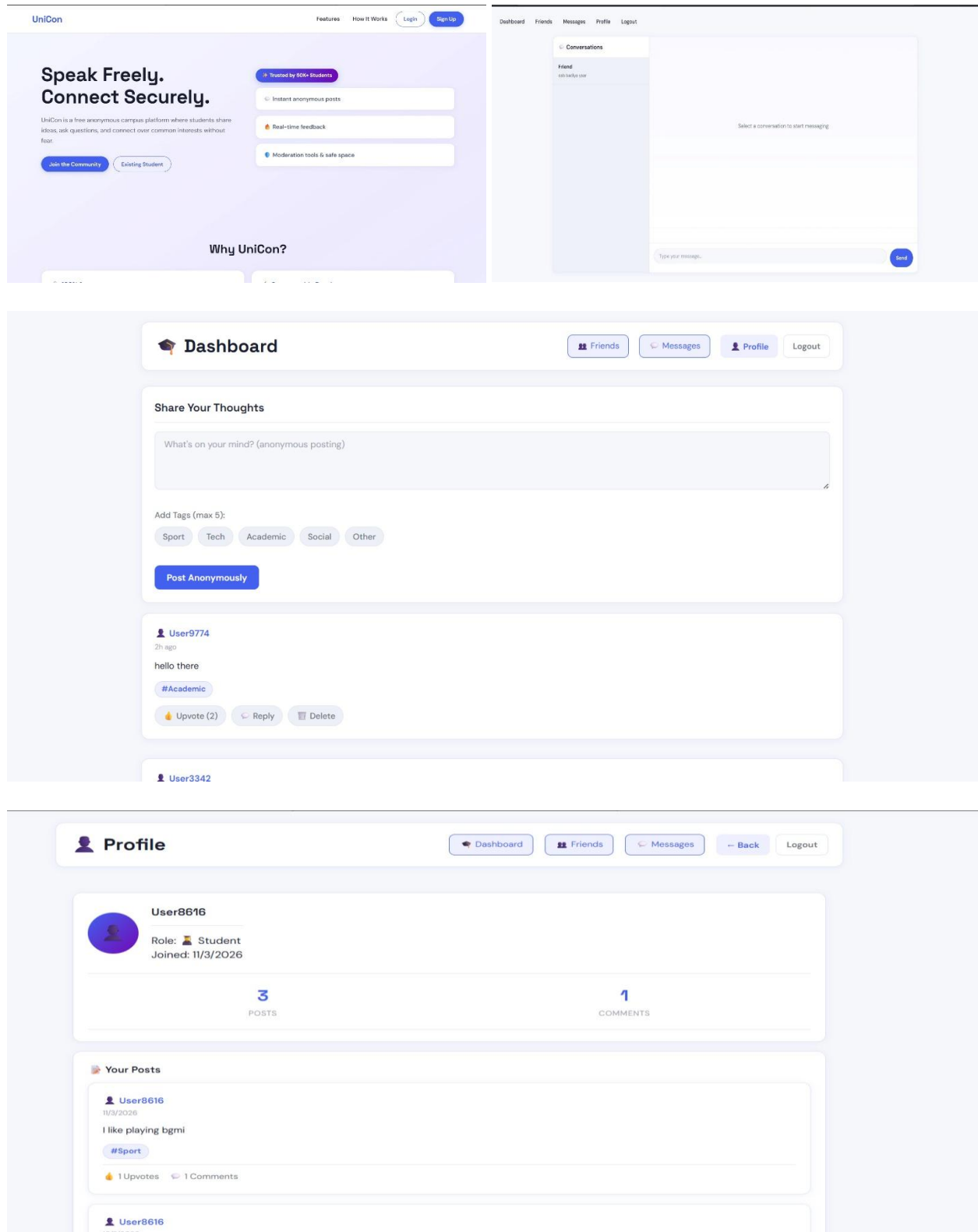


Figure 6:

Algorithms/Pseudo Code OF CORE FUNCTIONALITY:

BEGIN

OPEN UniCon Platform

DISPLAY Login / Register Page

INPUT user_choice

```
IF user_choice = "Register" THEN
    INPUT user_details
    CREATE new_user
    STORE user in database
    DISPLAY "Registration Successful"
END IF
```

INPUT username, password

```
IF valid credentials THEN
    DISPLAY Dashboard
```

WHILE user is logged in DO

```
    DISPLAY options:
    1. Create Post
    2. View Posts
    3. Interact (Like/Comment)
    4. Send Message
    5. Manage Friends
    6. View Profile
    7. Logout
```

INPUT user_action

```
IF user_action = Create Post THEN
    INPUT content, tag
    IF content is not empty THEN
        STORE post in database
        DISPLAY "Post Created"
    ELSE
        DISPLAY "Invalid Post"
    END IF
END IF
```

```

IF user_action = View Posts THEN
    FETCH posts from database
    DISPLAY posts
END IF

IF user_action = Interact THEN
    INPUT postID
    INPUT action (like/comment)
    STORE interaction
END IF

IF user_action = Send Message THEN
    INPUT receiverID, message
    IF message is not empty THEN
        STORE message in database
        DISPLAY "Message Sent"
    END IF
END IF

IF user_action = Manage Friends THEN
    INPUT friend_action (add/remove)
    UPDATE friend list in database
END IF

IF user_action = View Profile THEN
    FETCH user data
    DISPLAY profile details
END IF

IF user_action = Logout THEN
    EXIT loop
END IF

END WHILE

ELSE
    DISPLAY "Invalid Login"
END IF

END

```

7. PROJECT CLOSURE

7.1. Goals / Vision

The vision of UniCon is to provide a secure and anonymous platform where university students can freely express their thoughts, engage in discussions, and connect with peers without fear of judgment. The primary goal of the project is to enable open communication through features such as anonymous posting, tag-based content organization, private messaging, and a friend system, while ensuring a safe environment through moderation. The system is designed to be user-friendly, scalable, and adaptable for future enhancements, promoting a balanced ecosystem of privacy, interaction, and accountability within student communities.

7.2. Delivered Solution

The UniCon project was initially planned as an anonymous, campus-specific communication platform to address the lack of safe and open discussion spaces for students. The goal was to design a system that allows users to interact freely while maintaining privacy and ensuring accountability through moderation. The project aimed to include features such as anonymous profiles, structured discussions, and private communication.

The delivered solution is a fully functional web-based platform that successfully implements these planned features. It includes anonymous user interaction, post creation with tag-based categorization, commenting and engagement features, private messaging, and a friend system. Additionally, the system incorporates moderation controls to handle inappropriate behavior. The platform is supported by a modular architecture and a responsive user interface, ensuring smooth performance and ease of use. Overall, the delivered solution meets the initial objectives and provides a practical and scalable system for student interaction.

7.3. Remaining Work

Future improvements to UniCon include developing a mobile application for better accessibility, implementing advanced moderation features such as AI-based content filtering, and adding a notification system for real-time updates. Additionally, the platform can be expanded to support multiple universities and enhanced with performance optimizations and improved security measures for large-scale usage.

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